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VIT BHOPAL UNIVERSITY

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Library Management System

A CLI Based Application

Subject Code: CSE2006

Subject Name: Programming in Java

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PROJECT REPORT

Abstract

This project is a simple CLI app for college library. It helps librarian manage books and students issue and return books. Data is saved in CSV files. The app has login, book management, issue and return, and simple reports. It is made for easy understanding and runs with Java and Maven.

Table of Contents

1. Introduction	2
2. Problem Statement	2
3. Objectives	2
4. Scope	2
5. Target Users	3
6. Functional Requirements	3
7. Non-Functional Requirements	4
8. System Architecture	4
9. Process Flow	5
10. UML Diagrams	5
11. Database Design	6
12. Implementation	6
13. Testing	7
14. Screenshots	7
15. Conclusion	9
16. References	9

1. Introduction

College library was using manual register. It was slow and had errors. This project makes it digital with a simple menu system. It is made for college level, easy to run and understand. The project uses Java and Maven and stores data in CSV files.

2. Problem Statement

Manual tracking makes it hard to know which books are available, who has taken books, and when to return. Records get lost. Students wait and librarian does extra work. We need simple software to keep all this.

3. Objectives

- Make easy login and register
- Let admin add and manage books
- Let students search, issue and return books
- Show simple reports

4. Scope

In Scope

- Register and login
- Add, view, edit and delete books
- Search books

- Issue and return with limit check
- My books list
- Reports for admin
- CSV file storage

Out of Scope

- Payment
- Email or SMS
- Barcode
- Web or mobile app

5. Target Users

Admin (Librarian)

- **Goal:** Needs to add books, see stock, see who has books
- **Pain:** Lot of manual entries, no quick search

Member (Student)

- **Goal:** Find books, take and return easily
- **Pain:** Not knowing if book is there and forgetting due date

6. Functional Requirements

- FR1: Register new member
- FR2: Login and check role
- FR3: Admin can add member and librarian
- FR4: Admin can add book with id, title, author, copies
- FR5: Admin can edit and delete book
- FR6: Search books by title or author
- FR7: Issue book if copy is there and limit not crossed
- FR8: Return book and show fine if late
- FR9: Show my active books
- FR10: Show reports like total books, overdue, most borrowed

Input/Output Structure:

Input	Check	Output
username, password	check csv	login ok or error
book id, title, author, copies	save to books.csv	book added
username and book id for issue	check copy and limit	issued or error
username and book id for return	check active issue	returned or fine

7. Non-Functional Requirements

- **Performance:** works fast for 100 books, file read each time
- **Security:** role check, admin pages only for admin
- **Usability:** simple menu, clear messages
- **Reliability:** if csv missing it is created, bad line skipped
- **Maintainability:** separate packages for model, repo, service

- **Logging:** simple console output

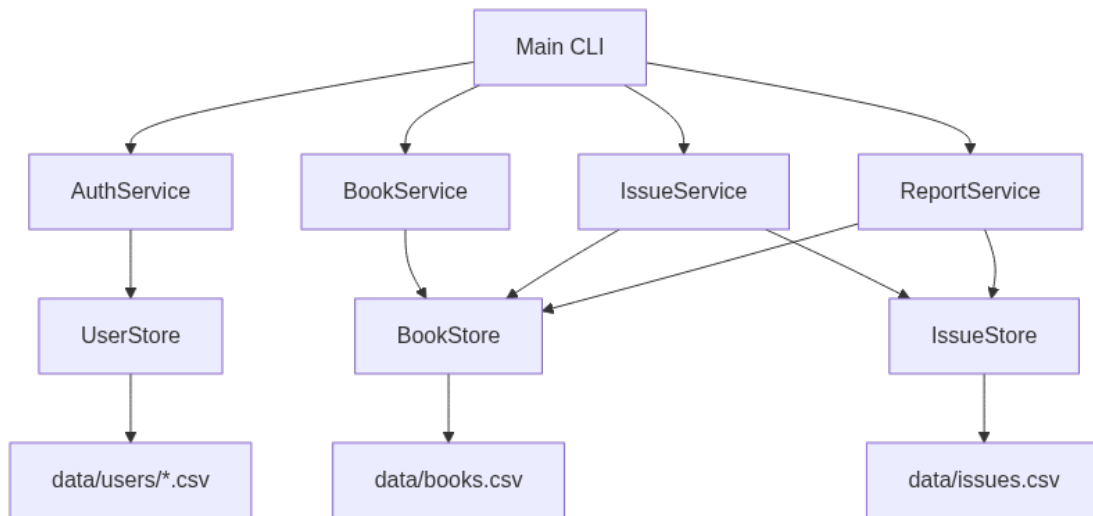
8. System Architecture

Modules: model, repo, service, util, Main. Each does one work. CSV files are used for storage, no database needed.

Folder structure:

```
src/main/java/com/cse2006/library/
  model/User, Book, Issue, Role
  repo/UserStore, BookStore, IssueStore
  service/AuthService, BookService, IssueService, ReportService
  util/Validator, Ansi
  Main
```

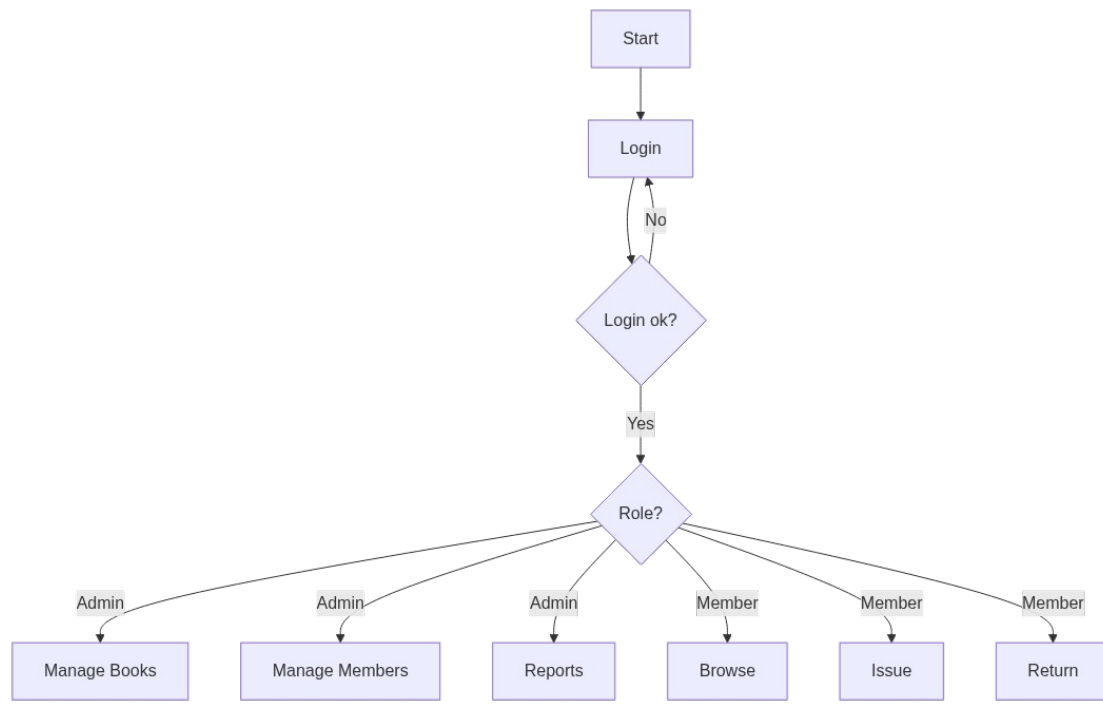
Architecture: Main CLI -> Services -> Repo -> CSV files in data/



Screenshot of graph TD diagram

9. Process Flow

Steps: start -> login/register -> go to dashboard based on role -> do work -> logout.



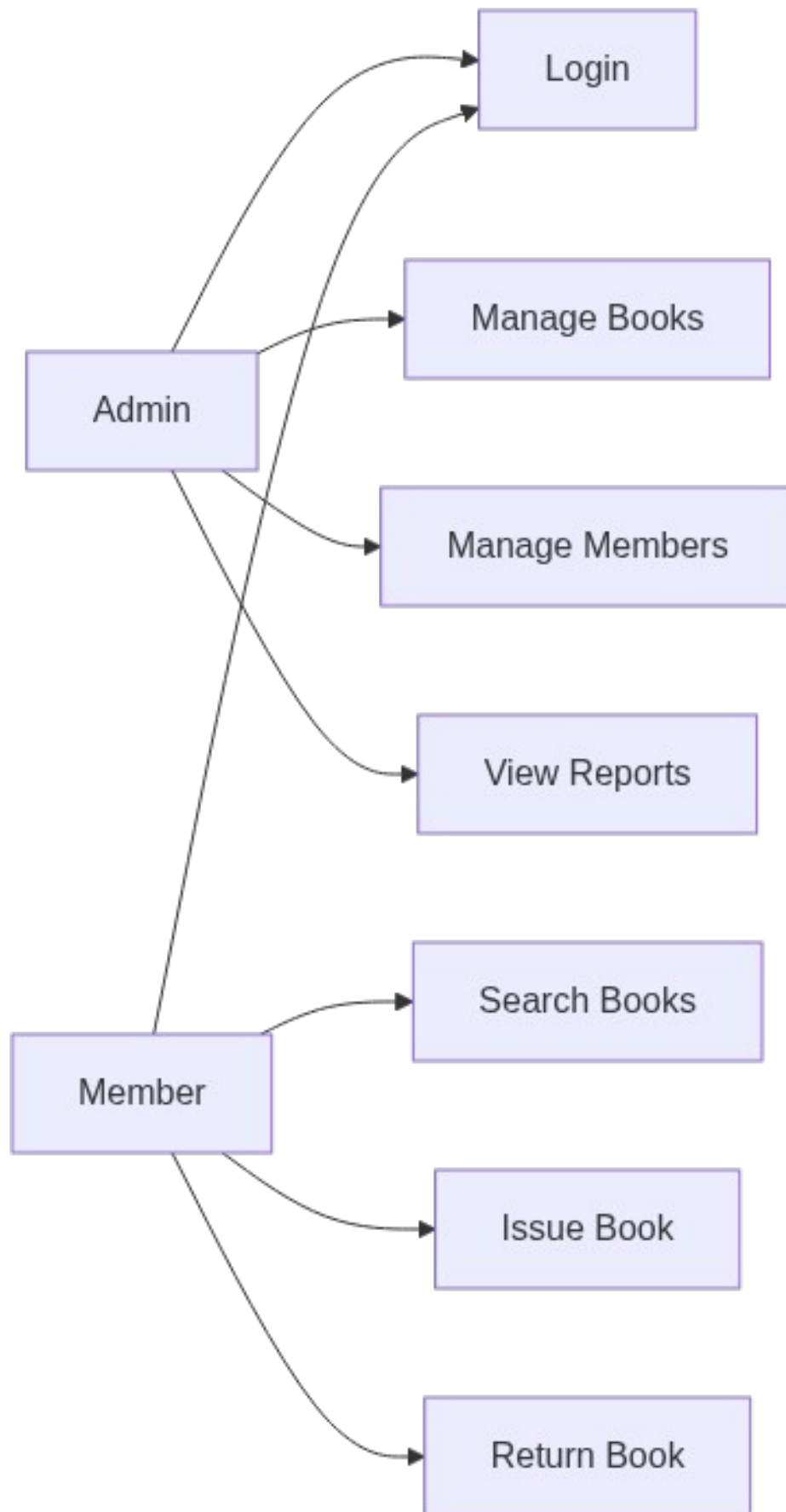
Screenshot of flowchart TD diagram

10. UML Diagrams

Use Case

Admin: Login, Manage Books, Manage Members, View Reports

Member: Login, Search Books, Issue Book, Return Book



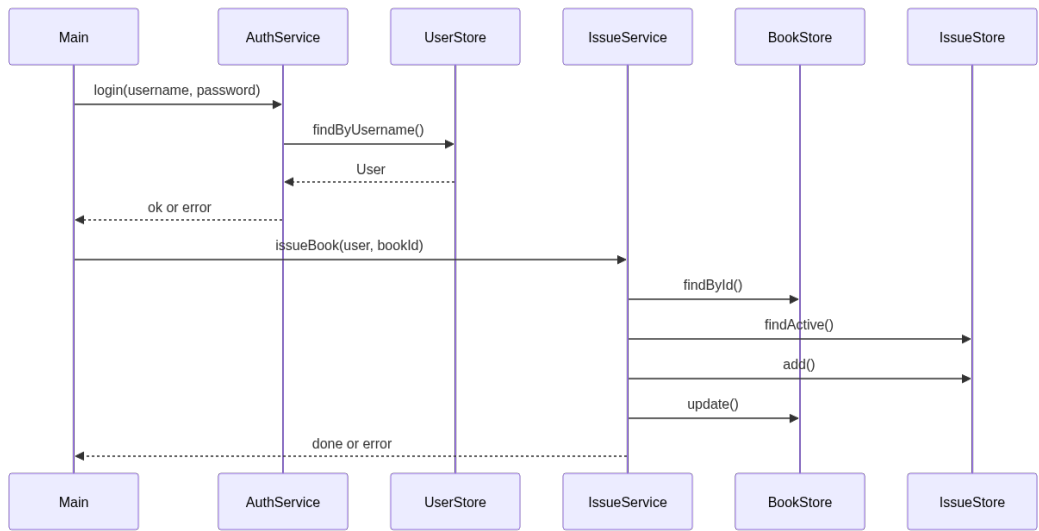
Screenshot of use case diagram

Class Diagram



Screenshot of classDiagram diagram

Sequence

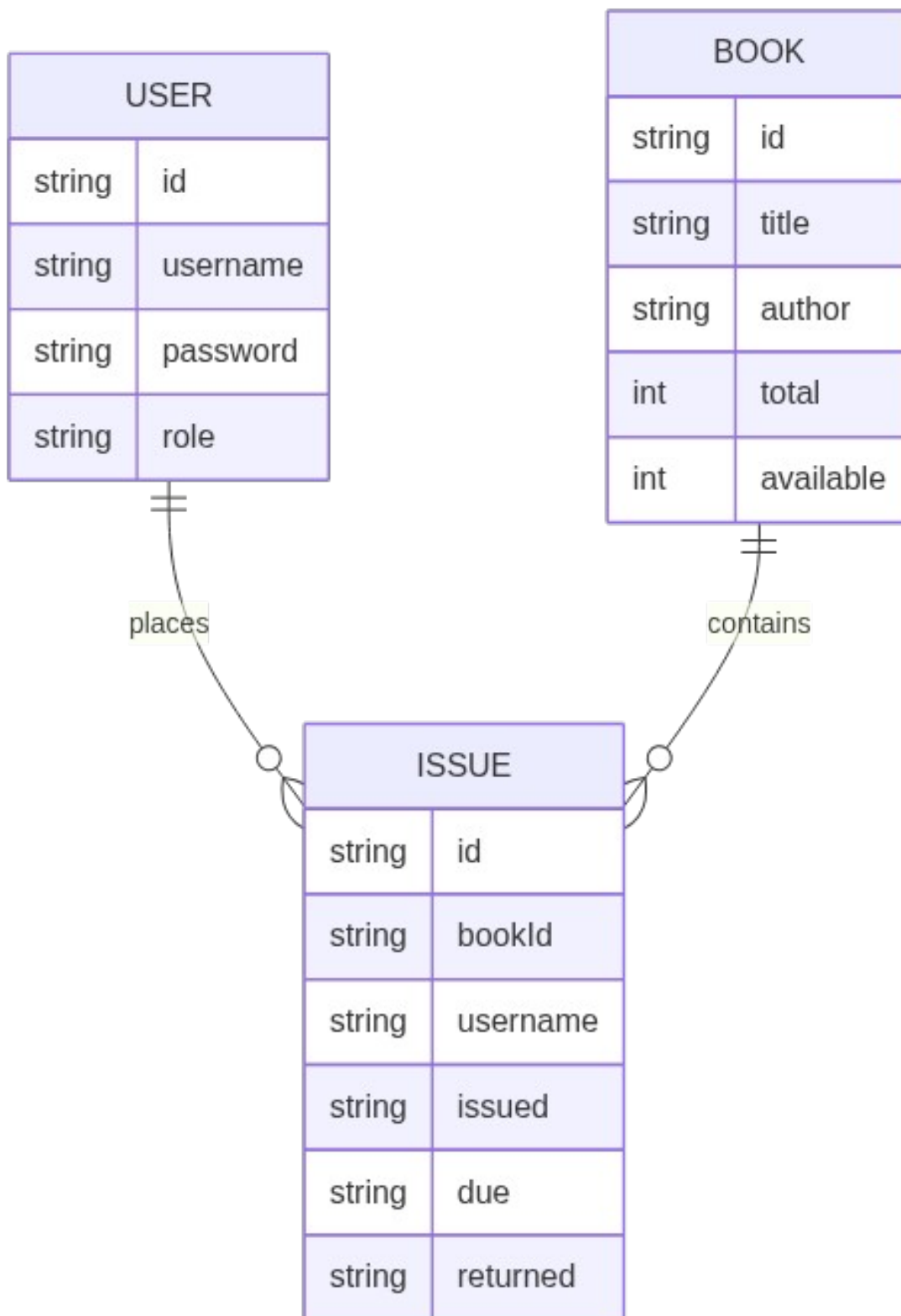


Screenshot of sequenceDiagram diagram

11. Database Design

Files in data folder, not pushed to github. Each user has own csv, books and issues are single files.

ER Diagram



Screenshot of erDiagram diagram

Schema

users: data/users/<username>.csv -> username,password,role,id,createdAt e.g.
admin,admin123,ADMIN,a1,2026-09-18

books: data/books.csv -> id,title,author,total,available e.g. B001,Intro to Java,Herbert Schildt,5,5

issues: data/issues.csv -> id,bookId,username,issued,due,returned e.g. abc123,B001,alice,2026-09-18,2026-10-02,

Rules: username is file name, book id is unique, issue is active if returned is empty

12. Implementation

Language: Java 17, Build: Maven, Storage: CSV files, CLI: Scanner and simple menu

Key code is simple if-else and file read/write, no framework. All logic is in service classes with plain checks.

To run:

```
git clone https://github.com/SykikX0/CSE2006.git
cd CSE2006
mvn compile
mvn exec:java -Dexec.mainClass="com.cse2006.library.Main"
# demo logins: admin/admin123 , alice/pass123
```

To test:

```
mvn test
```

13. Testing

Tests in src/test/java/com/cse2006/library/BasicTest.java:

- username and password check
- book add and find
- user register and login
- book service add/search/delete
- issue service issue and return
- report counts

Result: 7 tests, all pass. Run mvn test to verify.

14. Screenshots

All screenshots are taken from actual program run. They show the CLI menus and flows.

Login and Auth Menu - shows demo logins and member registration note

```
Terminal - Library Management System

+-----+
| Library Management System - CLI Auth |
+-----+

Demo: admin / admin123 | alice / pass123

1. Login
2. Register as member
3. About
4. Exit
Member registration here. Librarians are created by managers.

Select [1-4]: 1

+-----+
| Login      |
+-----+

Username: admin
Password: admin123
Welcome, admin (ADMIN)

Press enter to continue...
```

Admin Dashboard - manage books, members, reports

```
Terminal - Library Management System

+-----+
| Admin Dashboard - admin |
+-----+

1. Manage books
2. Manage members
3. View reports
4. List users (2 total)
0. Logout

Select 1
```

Manage Books - list with Avail/Total, add/search/edit/delete

```
Terminal - Library Management System

+-----+
| Manage Books - admin |
+-----+

1. List all books
2. Add book
3. Search books
4. Edit book
5. Delete book
0. Back

Select 1

ID Title Author Avail/Total
-----
B001 Intro to Java Herbert Schildt 5/5
B002 Clean Code Robert Martin 3/3
B003 Design Patterns GoF 2/2

Press enter to continue...
```

Member Dashboard - browse, search, issue, return, my books

```
Terminal - Library Management System

+-----+
| Member Dashboard - alice |
+-----+

1. Browse catalog
2. Search books
3. Issue book
4. Return book
5. My books
0. Logout

Select 3

Book ID to issue: B001
Issued Intro to Java due 2026-09-30

Press enter to continue...
```

Reports - totals, copies, issues, overdue and most borrowed

```
Terminal - Library Management System

+-----+
| Reports - admin |
+-----+

Books total: 3
Copies: 10 available / 10 total
Issues total: 1
Active issues: 0
Overdue: 0

Most borrowed: Intro to Java (B001) borrowed 1 times

Press enter to continue...
```

Issue and Return flow - issuing a book and viewing my books

```
Terminal - Library Management System

Your books:
- B001 Intro to Java due 2026-09-30
Book ID to return: B001
Returned on time

Press enter to continue...
```

15. Conclusion

The app covers all asked parts: user management, catalog and circulation, reporting, with simple CSV storage and clear menu flow. It is easy to run and check. Future work can add PDF export or fine payment. The code is kept simple for college level, easy to understand and explain.

16. References

- Java docs - docs.oracle.com/javase/17
- Maven docs - maven.apache.org
- College lab notes for library system - CSE2006